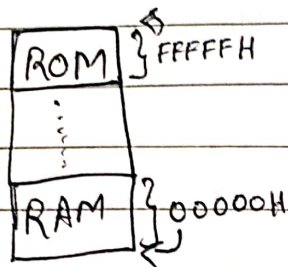


→ Memory Design

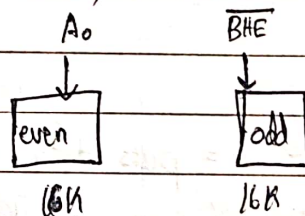


Design an 8086 maximum mode system with following Specifications 32KB using 16KB ROM Chips and 128KB RAM using 32KB RAM chips

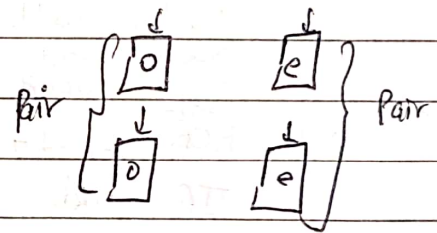
∴ For ROM :- $\frac{32KB}{16KB} \Rightarrow 2 \text{ chips}$

For RAM $\Rightarrow \frac{128KB}{32KB} \Rightarrow 4 \text{ chips}$

one will be even, one will be odd



two will be even, two will be odd



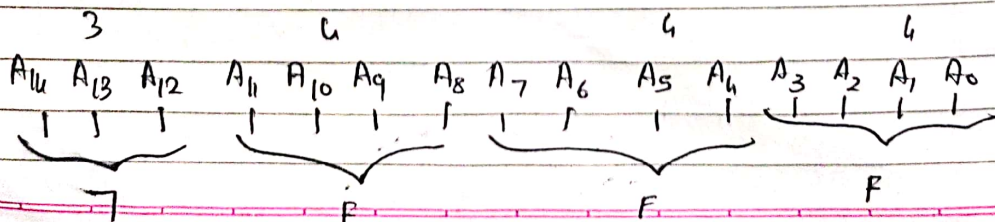
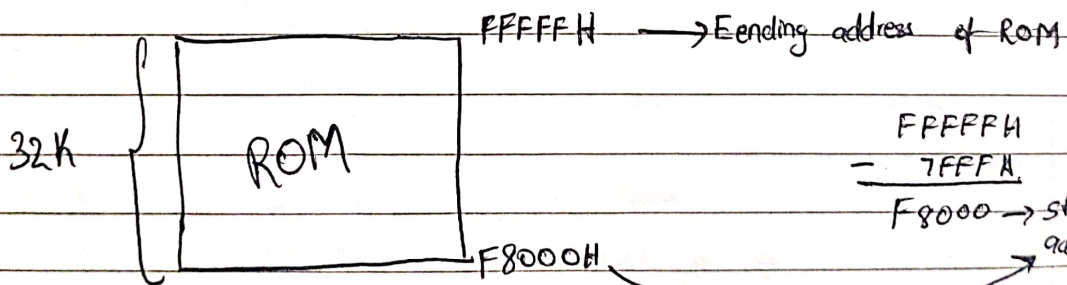
$$2^{10} \times 2^9 \Rightarrow 32K$$

$$\downarrow \quad \downarrow$$

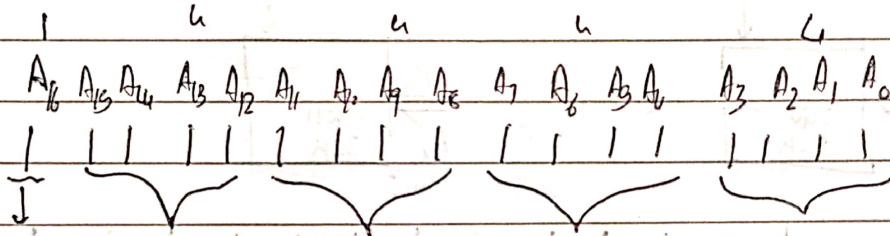
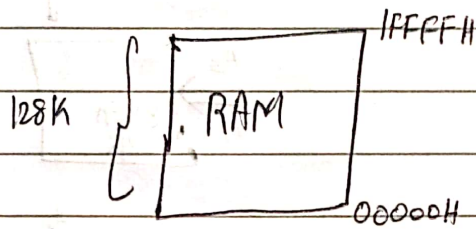
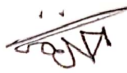
$$1K \quad 32$$

∴ Summation of the powers gives total number of address lines = $10+5 = 15 \text{ lines}$

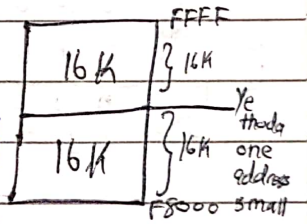
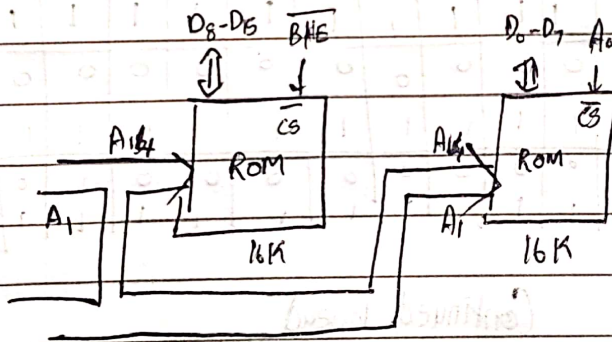
$A_0 - A_{14}$



Teacher's Signature:



$$\begin{array}{r} 00000 \\ + 1FFFF \\ \hline 1FFFFH \end{array}$$



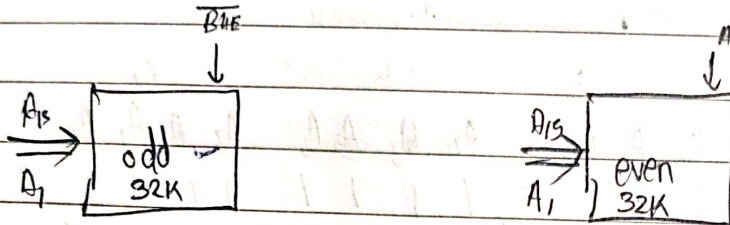
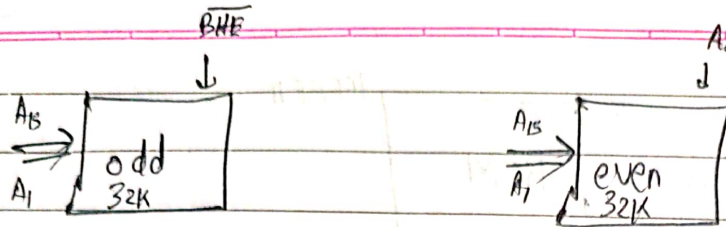
A1-A16
These will be all 0 or all 1

Starting address

	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
Even	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	F8000H
	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFFH
Odd	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	F8001H
	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	FFFFH

↑
ending address

Whistle podu



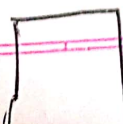
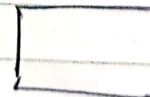
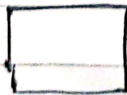
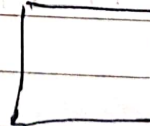
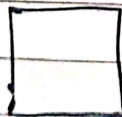
	A ₁₇	A ₁₆	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
$\overline{Y_0}$ {	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0000H
	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0FFFH
$\overline{A_0}$ {	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0001H
	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0FFFFH
$\overline{Y_1}$ {	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1000H
	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1001H
$\overline{A_1}$ {	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1FFFFH
	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1FFFFH

Continued Ahead

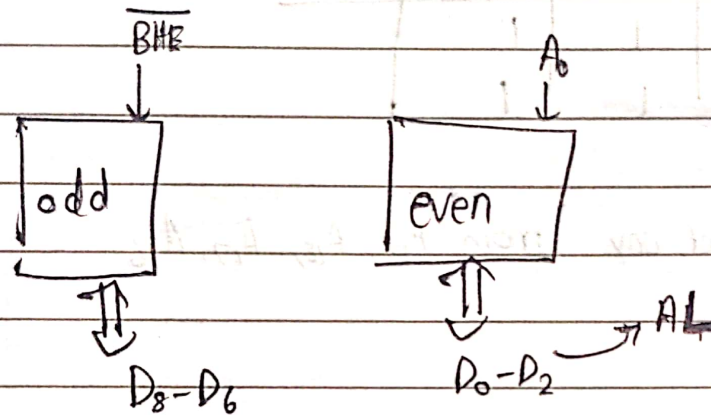
Q) Og size 128kB
chip size 16kB

$$\therefore \text{No of chips} = 8 \rightarrow \frac{128}{16} = 8$$

$$\text{Address lines used} = 10 \times 2^4 = 16 \text{ lines}$$



Teacher's Signature:



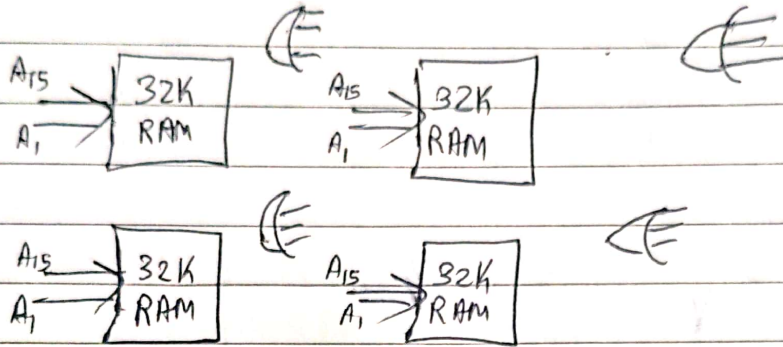
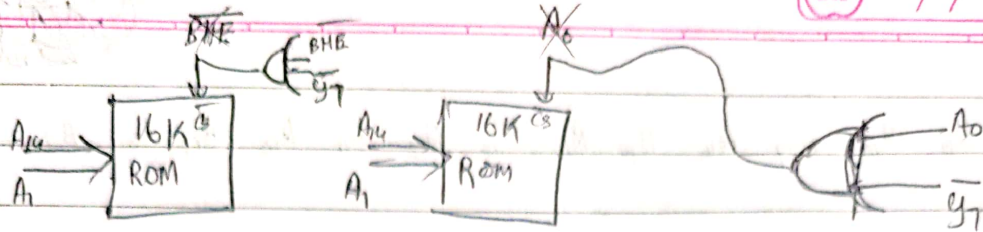
$AH \leftarrow$

`MOV AX, [0000]`

↑
if this is
16 bit

→ If this is odd then 2 clock cycles
if even toh 1 clock cycle only

Teacher's Signature:



A_1 to A_{14} is used for ROM

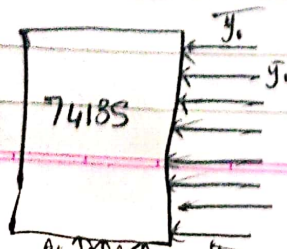
A_{19}	A_{18}	A_{17}	A_{16}	A_{15}	A_{14} $\longleftrightarrow$ A_7
1	1	1	1	1	
1	1	1	1	1	
1	1	1	1	1	
1	1	1	1	1	

Don't select A_{15} , select any from $A_{19}, A_{18}, A_{17}, A_{16}$
 $\downarrow$
 used in RAM

Upper lines are constant

But we can only connect single line to chips

Decoder is used



If you're not connecting lines to decoder then partial decoding

Teacher's Signature:

Maximise $Z = 8x_1 + 7x_2 - 2x_1^2 - x_2^2$
 Subject to $4x_1 + 3x_2 = 22$
 $3x_1 + x_2 = 14$
 $x_1, x_2 \geq 0$

So we can take

4 3 1

then

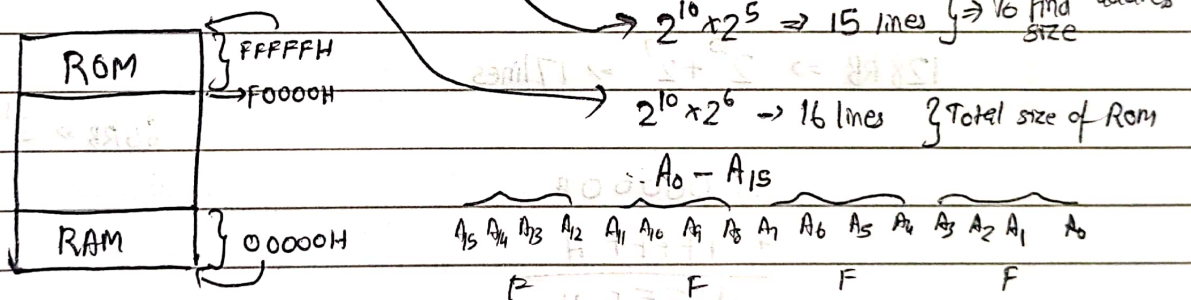
22 14 0

→ Practice problems

Q Design 8086 system in minimum mode with the following specifications
 64Kb rom using 32Kb rom
 128Kb RAM using 64Kb RAM

A Step 1: $\frac{64}{32} \Rightarrow 2$ ROM chips
 $\frac{128}{64} \Rightarrow 2$ RAM chips

Draw entire Minimum mode Architecture first

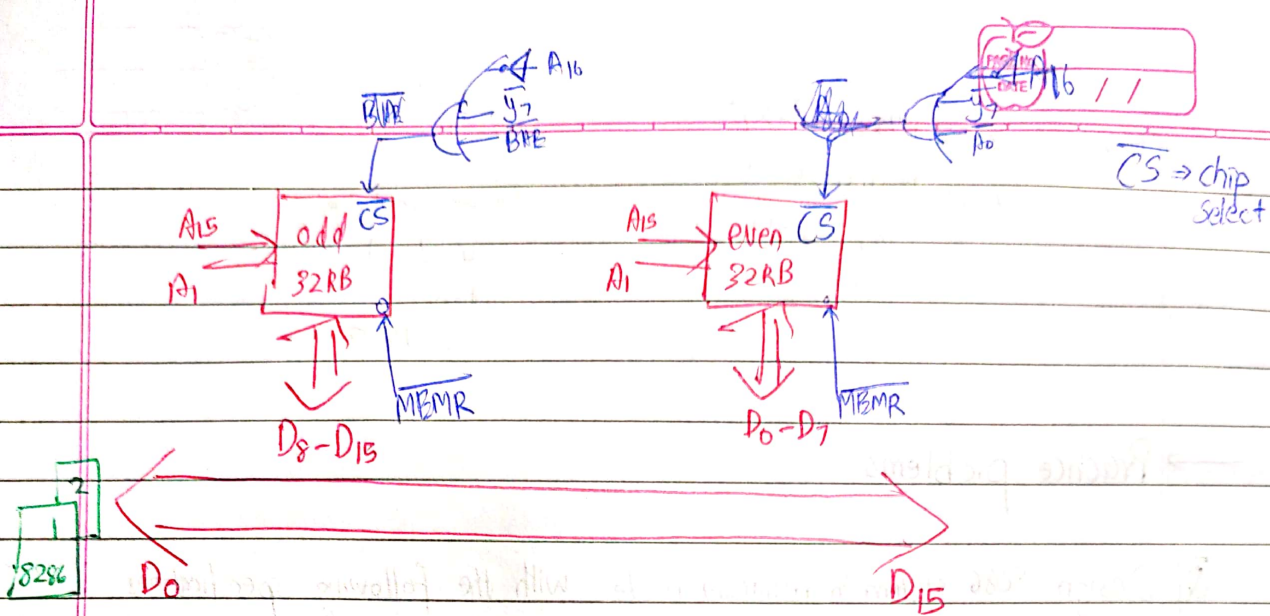


Step 2:

This gives starting address of ROM

$$\begin{array}{r} FFFFFH \\ - 0FFFFH \\ \hline F0000H \end{array}$$

	A_{15}	A_{14}	A_{13}	A_{12}	A_{11}	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0
$F0000H$ even	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
$FFFFFH$ odd	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
$F0001H$ even	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1
$FFFFFH$ odd	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1



Mov BX, [1000] $\rightarrow$ even ; if it was 1001 then odd
 $\uparrow$ Both chips are selected
 if BL or BH, then one chip

Now for RAM

00000H

128KB $\Rightarrow 2^{10} + 2^7 \Rightarrow 17$ lines

64KB $\Rightarrow 2^{10} + 2^6 \Rightarrow 16$ lines

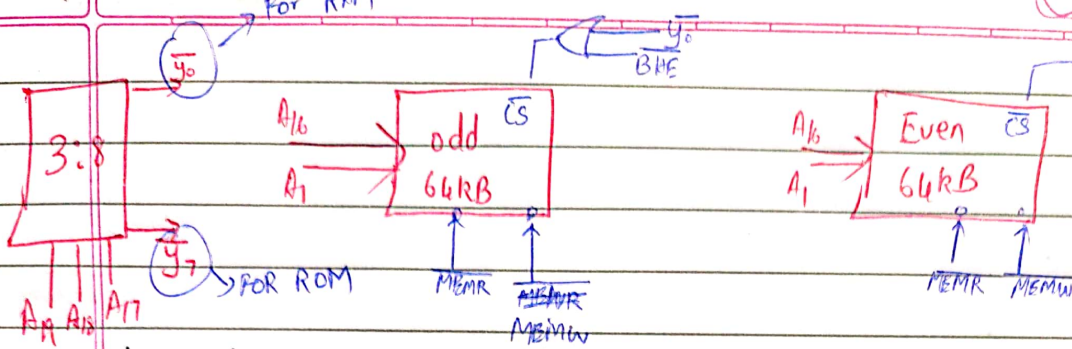
00000H

+ 1FFFFH

1FFFFH

	A ₁₉	A ₁₈	A ₁₇	A ₁₆	A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀
00000H	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
even	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1FFFFH	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
00001H	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
1FFFFH	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

C₁₆ A₁₇ A₁₈ A₁₉ is 000 for RAM



In the decoder you have to take common constant address lines i.e. A₁₇, A₁₈, A₁₉

We want the output of OR^{gate} to be 0

Also utilize all lines

- Q Design a maximum mode memory system using the following specifications
- 64KB ROM using 16KB ROM chip
 - 16KB RAM using 8KB RAM chip